Technical Document #60: Machine Learning - Comprehensive Overview

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Feature selection is the process of selecting a subset of relevant features for use in model construction. It helps reduce overfitting and improves model interpretability.

Neural networks are computing systems inspired by biological neural networks. They consist of layers of interconnected nodes that process information using connectionist approaches.

Overfitting occurs when a model learns the training data too well, including noise and outliers, leading to poor generalization on unseen data. Regularization techniques help prevent this.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Gradient descent is an optimization algorithm used to minimize the cost function in machine learning models.

- Cross-validation is a statistical method used to estimate the skill of machine learning models.

- Ensemble methods combine multiple machine learning models to create a more powerful model.